

SmartInternz / SkillWallet

Project Report

Library Management System using ServiceNow

Submitted by

Pakki Bonisha (Team Lead)

Date: [15-07-2026]

Date	[15-07-2026]
Team ID	0
Project Name	Library Management System using ServiceNow
Team Lead	Pakki Bonisha
Team Members	Teegala Divya Sri, Bhargava Pinapathruni, Prakash Potnuru, Sreemayee Gudla
Mentor	N-A
Skills Required	ServiceNow, Workflow Management, Access Control List, Users, Groups & Roles Management

1. Introduction

1.1 Project Overview

This project involves developing a role-based Library Management System on the ServiceNow platform to automate and streamline library operations in an academic environment. The system enables students to request books and track request status, while librarians manage the book inventory and control the approval process. Role-based access controls ensure secure operations, automated workflows handle request processing and availability updates, and reporting features provide insights into borrowing trends. The solution improves visibility, efficiency, and accuracy in managing library resources.

Two primary roles drive the system: Students, who browse available books and submit borrow requests, and Librarians, who review, approve, issue, and track the return of books. Every request moves through a defined lifecycle — Requested → Approved → Issued → Returned — with automated notifications and validations at each stage, ensuring transparency and reducing manual overhead for library staff.

1.2 Purpose

The purpose of this project is to design and implement an end-to-end library automation solution on ServiceNow that:

- Eliminates manual, error-prone tracking of book inventory and borrow requests.
- Enforces role-based responsibilities so students and librarians only see and act on what is relevant to them.
- Automates the approval, issuance, and return workflow using Flow Designer, reducing turnaround time for requests.
- Provides real-time visibility into book availability and request status.
- Generates actionable reports (e.g. Most Borrowed Books, Active Book Requests) to support library decision-making.

2. Ideation Phase

2.1 Problem Statement

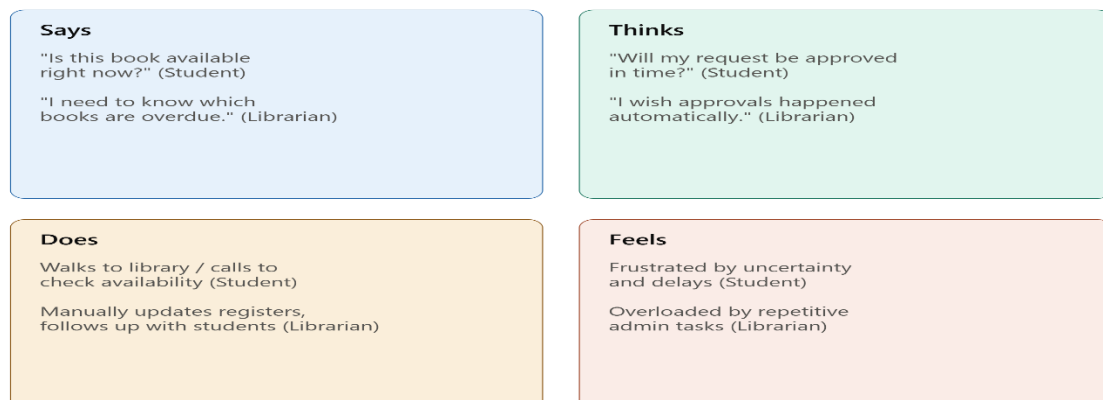
Traditional library management in academic institutions is largely manual — book issue/return registers, verbal or paper-based approval, and no centralized way to check book availability before visiting the library. This leads to duplicate requests for the same book, delayed approvals, lost track of overdue returns, and no analytical visibility for library staff into borrowing trends.

“As a student, I want to check book availability and request a book online, and as a librarian, I want to approve/reject and track issued and overdue books efficiently, without relying on manual registers.”

2.2 Empathy Map Canvas

The empathy map below captures the perspective of the two primary personas — Student and Librarian — used to shape the solution design.

Says	Thinks	Does	Feels
“Is this book available right now?” (Student)	“Will my request be approved in time for my exam?” (Student)	Walks to the library / calls to check availability (Student)	Frustrated by uncertainty and delays (Student)
“I need to know which books are overdue.” (Librarian)	“I wish approvals and status updates happened automatically.” (Librarian)	Manually updates registers and follows up with students (Librarian)	Overloaded by repetitive administrative tasks (Librarian)



2.3 Brainstorming

Ideas generated and prioritized during the ideation phase:

- Self-service portal for students to view real-time book availability.
- Automated librarian approval flow instead of manual sign-off.
- System-enforced validation so a return date is mandatory only once a book is issued.

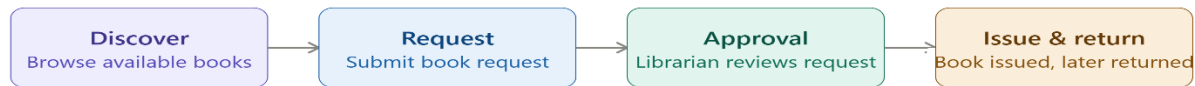
- Reference qualifier to show only currently available books when a student submits a request.
- Automated email notifications on approval and issuance.
- Built-in reporting for borrowing trends (Most Borrowed Books, Active Requests).



3. Requirement Analysis

3.1 Customer Journey Map

Stage	Discover	Request	Approval	Issue & Return
Student Action	Logs in to portal, browses available books	Submits a Book Request for a chosen title	Waits for librarian approval/rejection notification	Collects issued book; returns it by the due date
Librarian Action	Maintains catalog and availability status	Receives the request in queue	Reviews and approves/rejects the request	Marks book as Issued, later as Returned; updates availability



3.2 Solution Requirement

Functional Requirements:

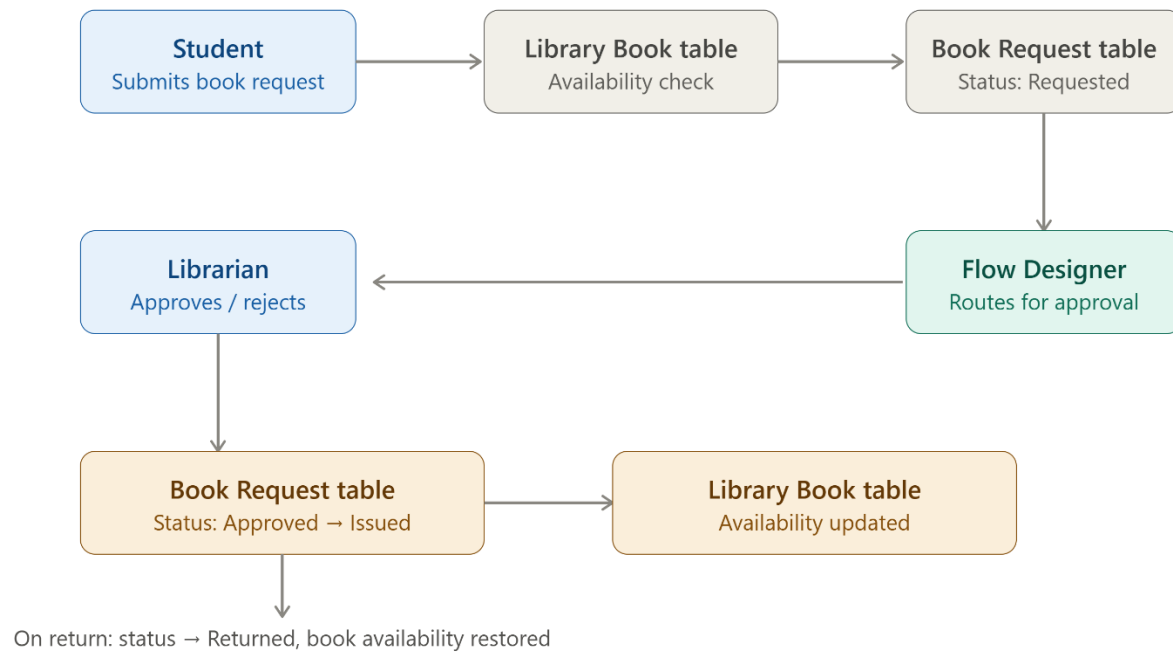
FR No.	Functional Requirement (Epic)	Sub Requirement (Story)
FR-1	Role Management	Create Student and Librarian roles; assign to respective users
FR-2	Book Catalog Management	Create Library Book table (Book ID, Title, Author, Category, Availability Status, Total Copies)
FR-3	Book Request Management	Create Book Request table (Requested Book, Requested By, Request Status, Request Date, Return Date)
FR-4	Access Control	ACLs so students can view/request books; librarians can create, update, approve, reject
FR-5	Approval Workflow	Flow Designer flow: request submission → librarian approval → status update → issue → return
FR-6	UI Validation	UI Policy making Return Date mandatory only when a book is issued
FR-7	Availability Automation	Automatic update of book availability on approval and on return
FR-8	Reporting	Reports: Most Borrowed Books, Active Book Requests

Non-Functional Requirements:

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple portal navigation: Home → Library Services → Submit Borrow Request
NFR-2	Security	Role-based ACLs restrict data access and actions by role
NFR-3	Reliability	Reference qualifier prevents requests for unavailable books
NFR-4	Performance	Flow Designer executes approval/issue/return updates in real time
NFR-5	Availability	System accessible to students and librarians throughout working hours

3.3 Data Flow Diagram

The data flow begins when a student submits a Book Request from the portal. The request is validated against book availability (via a reference qualifier), routed to the librarian through Flow Designer for approval, and on approval the Book Request status and the Library Book availability are updated in sequence. Issue and return actions further update both tables and trigger notifications.



3.4 Technology Stack

S.No	Component	Description	Technology
1	Platform	Core application platform hosting all custom tables and logic	ServiceNow PDI
2	User Interface	Service Portal for students and Librarian workspace UI	ServiceNow Service Portal / Now UI
3	Data Model	Custom tables storing books and borrow requests	ServiceNow Table Builder (u_book, u_book_request)
4	Business Logic / Workflow	Approval, issue and return automation	Flow Designer
5	Client-side Validation	Mandatory Return Date only once a book is issued	UI Policy
6	Security	Role-based read/write/create access	Access Control Rules (ACL)
7	Notifications	Email alerts on approval and issuance	ServiceNow Notifications (Flow Designer action)
8	Reporting	Analytics on borrowing activity	ServiceNow Reporting module

4. Project Design

4.1 Problem - Solution Fit

Problem: Manual book borrowing has no visibility into availability, no structured approval trail, and no analytics for library staff.

Solution Fit: ServiceNow's table-driven data model, Flow Designer automation, and native reporting directly address these gaps — giving students self-service visibility and librarians a controlled, auditable approval and issuance process.

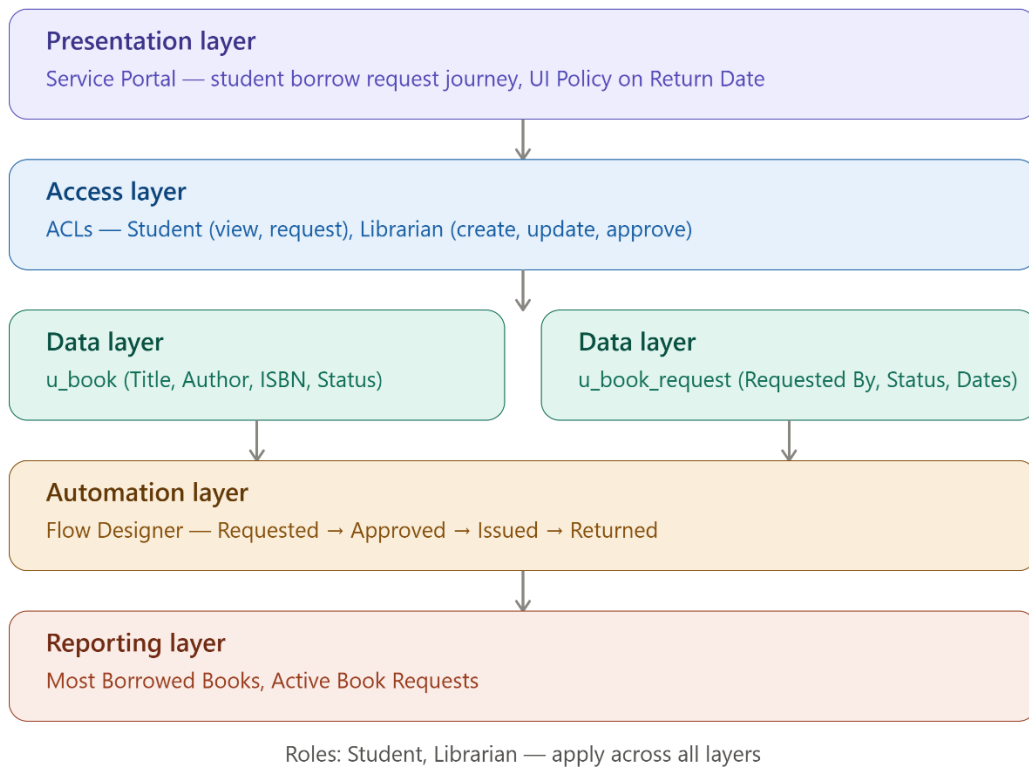
4.2 Proposed Solution

S.No	Parameter	Description
1	Problem Statement	Manual, register-based library borrowing with no real-time availability or automated approval
2	Idea / Solution description	A ServiceNow-based portal where students request books and librarians approve/issue/track returns through an automated Flow Designer workflow
3	Novelty / Uniqueness	Reference qualifier restricts requests to available books only; UI Policy enforces mandatory return date at the correct stage
4	Social Impact / Customer Satisfaction	Faster approvals and clear status tracking improve the borrowing experience for students and reduce administrative load for librarians
5	Business Model	Internal academic service; scalable to a licensed ServiceNow application for institutional deployment
6	Scalability of the Solution	Extendable to overdue notifications, external catalog integration, and other academic services (lab equipment, study room booking)

4.3 Solution Architecture

The architecture follows a role-driven, workflow-orchestrated model on the ServiceNow platform:

- Roles: Student and Librarian roles govern access throughout the application.
- Data layer: Library Book table (Book ID, Title, Author, Category, Availability Status, Total Copies) and Book Request table (Requested Book, Requested By, Request Status, Request Date, Return Date).
- Access layer: ACLs scope student access to viewing/requesting books, and librarian access to managing records and approvals.
- Automation layer: Flow Designer flow triggers on request submission, routes to the librarian, and updates status through Requested → Approved → Issued → Returned.
- Presentation layer: UI Policy enforces mandatory Return Date only once Issued; Service Portal provides the student-facing borrow request journey.
- Reporting layer: Most Borrowed Books and Active Book Requests reports for library analysis.



5. Project Planning & Scheduling

5.1 Project Planning

Sprint	Functional Requirement (Epic)	Story No.	User Story / Task	Story Points	Priority	Duration
Sprint-1	Role & Access Setup	USN-1	As an admin, I can create Student and Librarian roles and assign them to users	2	High	-
Sprint-1		USN-2	As an admin, I can create the Library Book and Book Request tables with required fields	3	High	-
Sprint-1	Access Control	USN-3	As an admin, I can configure ACLs so students view/request books and librarians manage records	2	High	-
Sprint-2	Workflow Automation	USN-4	As a librarian, I want requests routed to me automatically for approval via Flow Designer	3	High	-

Sprint-2	UI Validation	USN-5	As a librarian, Return Date becomes mandatory only once a book is Issued	1	Medium	-
Sprint-2	Reporting	USN-6	As a librarian, I can view Most Borrowed Books and Active Book Requests reports	2	Medium	-
Sprint-3	Documentation	USN-7	As a team, we provide a step-by-step manual to recreate the solution in a new PDI, with screenshots and order of operations	2	Medium	10m
Sprint-3	Demo Planning	USN-8	As a team, we plan a demo video covering portal walkthrough, live request, approval, notifications, and status tracking	2	Medium	20m
Sprint-3	Visual Demonstration	USN-9	As a team, we capture form behaviour, approval timeline, and flow logs with annotated visuals	1	Low	10m
Sprint-3	Explanation Clarity	USN-10	As a team, documentation follows a What → Why → How → Outcome structure with labelled screenshots	2	Low	20m

Development Steps Followed:

- Logged in to the ServiceNow instance using administrator credentials.
- Created Student and Librarian roles to separate access and responsibilities.
- Created users and assigned the appropriate roles to each.
- Created the Library Book table (Book ID, Title, Author, Category, Availability Status, Total Copies).
- Created the Book Request table (Requested Book, Requested By, Request Status, Request Date, Return Date).
- Configured ACLs for student view/request access and librarian create/update/approve access.
- Built a Flow Designer flow that triggers on request submission, routes to the librarian, and updates status.
- Configured automatic book availability updates on approval and on return.
- Created Most Borrowed Books and Active Book Requests reports.
- Tested the application by impersonating a student to submit a request and a librarian to approve/reject it.

- Verified role-based access, workflow execution, and data updates.
- Documented tables, roles, ACLs, workflows, and reports for reproducibility in a new PDI.

6. Functional and Performance Testing

6.1 Performance Testing

Testing focused on validating the borrow/return lifecycle end-to-end and confirming realistic, real-world library scenarios:

- Role-based approvals verified: student requests routed correctly to the librarian for approval.
- Data validations confirmed: book availability check prevents requesting an already-borrowed book; Return Date is enforced as mandatory only once a book is issued.
- Tested with realistic use cases, including multiple students requesting the same book and overdue return scenarios.
- Test data submission performed to validate the full borrow/return flow, with screenshots capturing the order of operations.
- Impersonation testing carried out for both Student and Librarian roles to confirm role-based access boundaries.

Evidence captured from PDI testing:

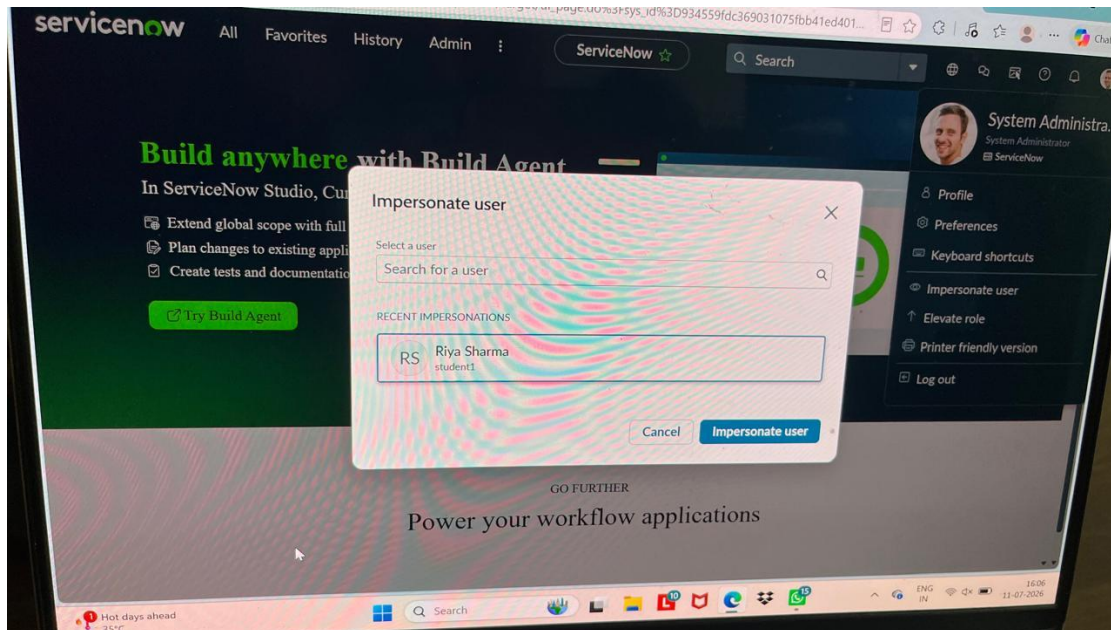


Fig 6.1 — Impersonate User dialog confirming impersonation as student user (Riya Sharma / student1), used to test role-based access.

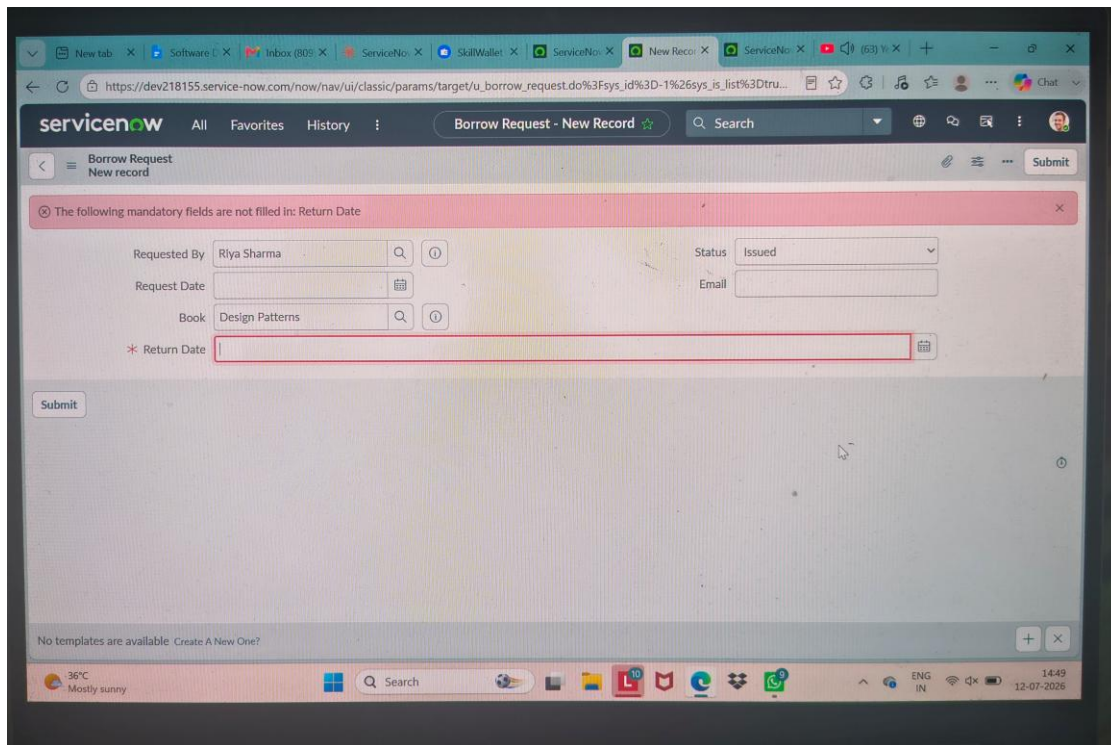


Fig 6.2 — UI Policy validation: system blocks Submit with "The following mandatory fields are not filled in: Return Date" when Status = Issued, confirming Return Date becomes mandatory only once a book is issued.

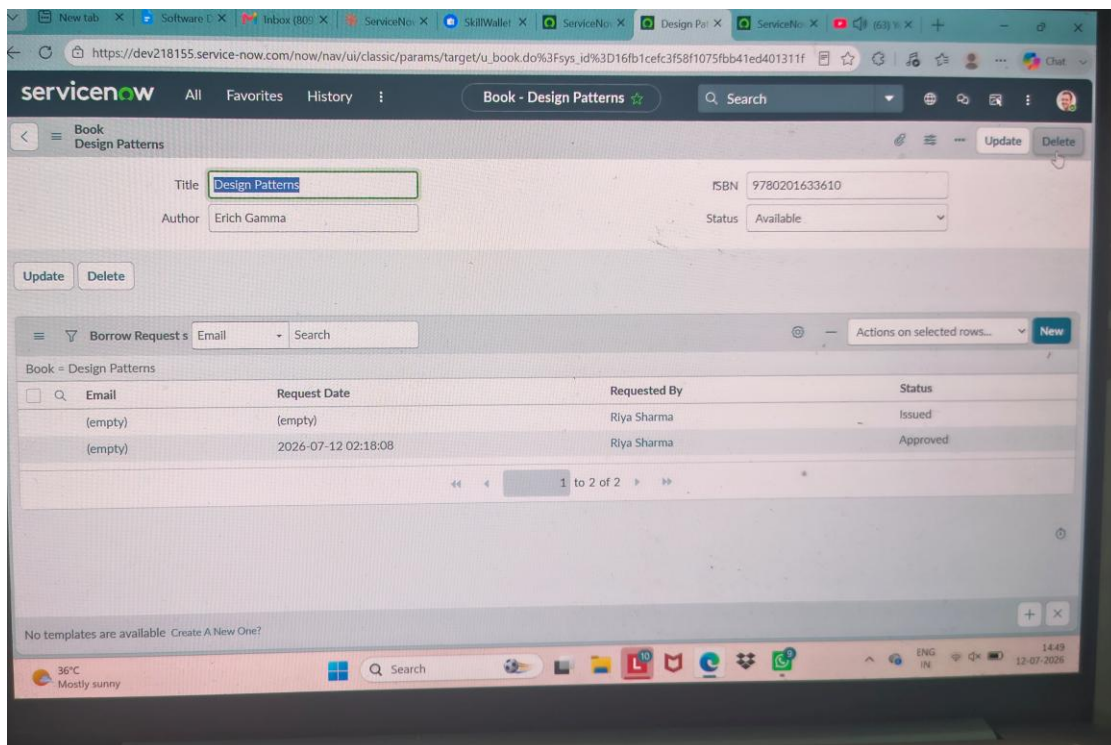


Fig 6.3 — Library Book record ("Design Patterns") with related Borrow Requests showing Issued and Approved statuses, confirming book-to-request relationship and availability tracking.

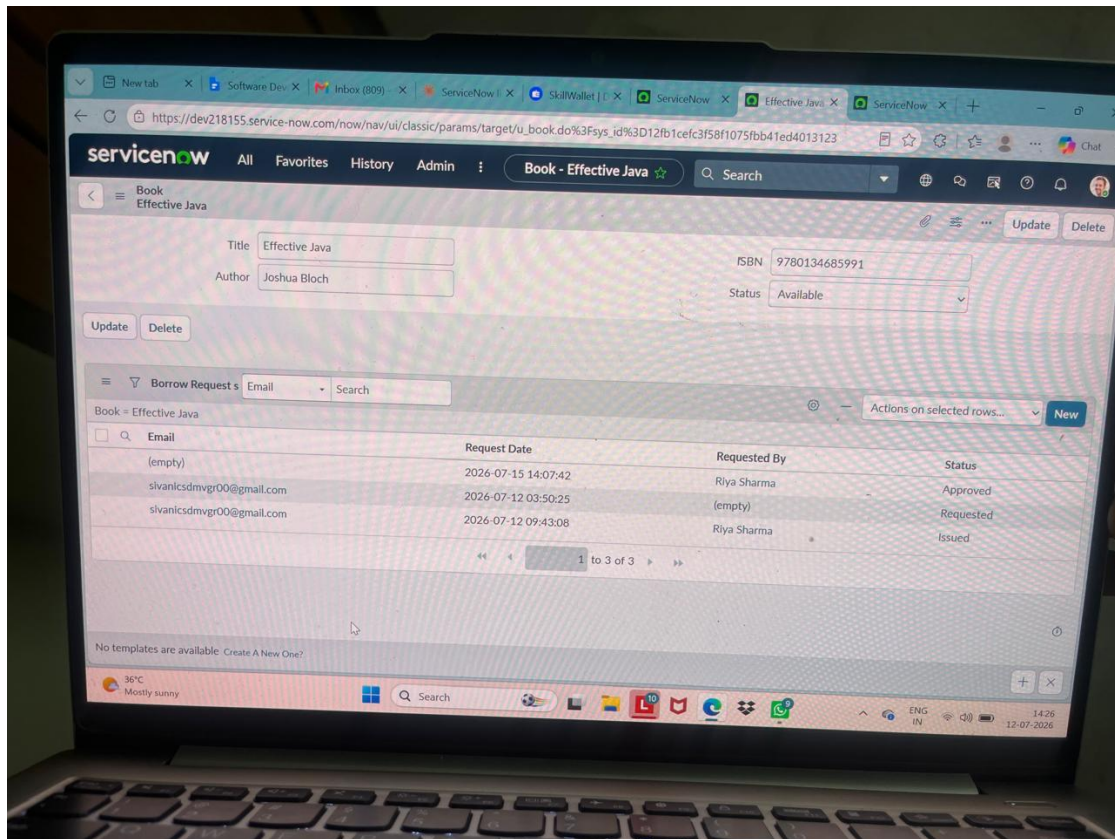


Fig 6.4 — Library Book record ("Effective Java") with related Borrow Requests (Approved, Requested, Issued), used to validate multiple requests against a single book.

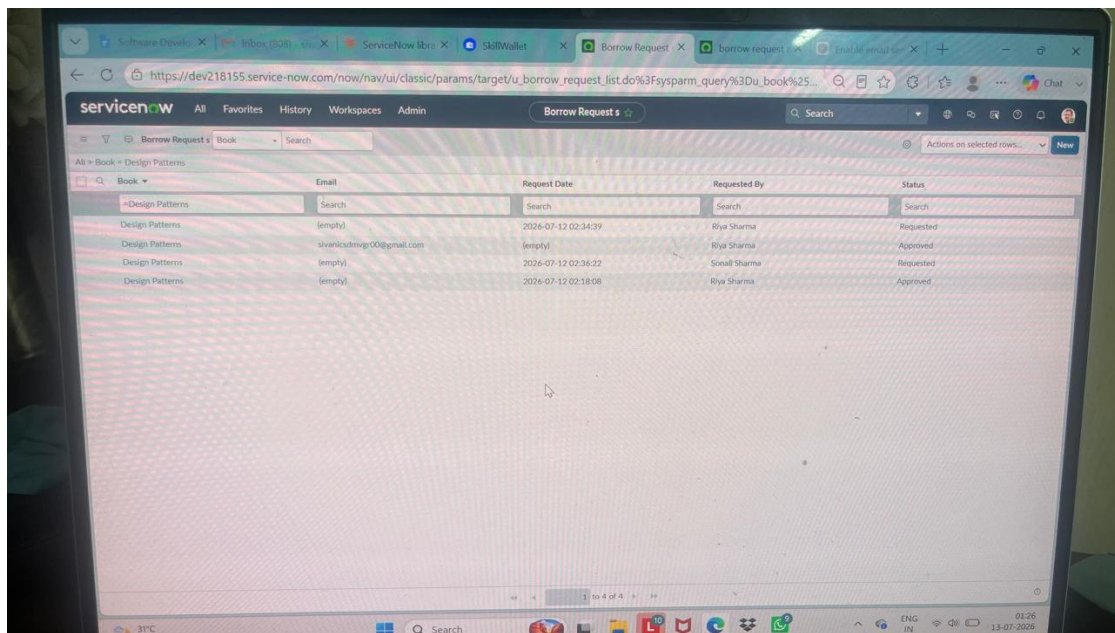


Fig 6.5 — Borrow Request list filtered by Book = "Design Patterns", showing multiple students requesting the same book (realistic test scenario) with different outcomes (Requested, Approved).

Book	Email	Request Date	Requested By	Status
Design Patterns	(empty)	2026-07-12 02:34:39	Riya Sharma	Requested
Effective Java	sivanicsdmvgr00@gmail.com	2026-07-12 03:50:25	Riya Sharma	Requested
The Alchemist	(empty)	2026-07-06 04:07:33	(empty)	Requested
Effective Java	(empty)	2026-07-15 14:07:42	Riya Sharma	Approved
First love	sivanicsdmvgr00@gmail.com	2026-06-30 03:40:45	System Administrator	Requested
Design Patterns	(empty)	(empty)	Riya Sharma	Issued
Design Patterns	(empty)	2026-07-12 02:36:22	Sonali Sharma	Requested
bujiima	(empty)	2026-07-14 03:03:49	Riya Sharma	Rejected
Design Patterns	(empty)	2026-07-12 02:18:08	Riya Sharma	Approved
Effective Java	sivanicsdmvgr00@gmail.com	2026-07-07 09:43:08	Riya Sharma	Issued

Fig 6.6 — Full Borrow Request test data list (10 records) showing the complete range of statuses exercised during testing: Requested, Approved, Issued, and Rejected — confirming the end-to-end borrow/return flow and order of operations.

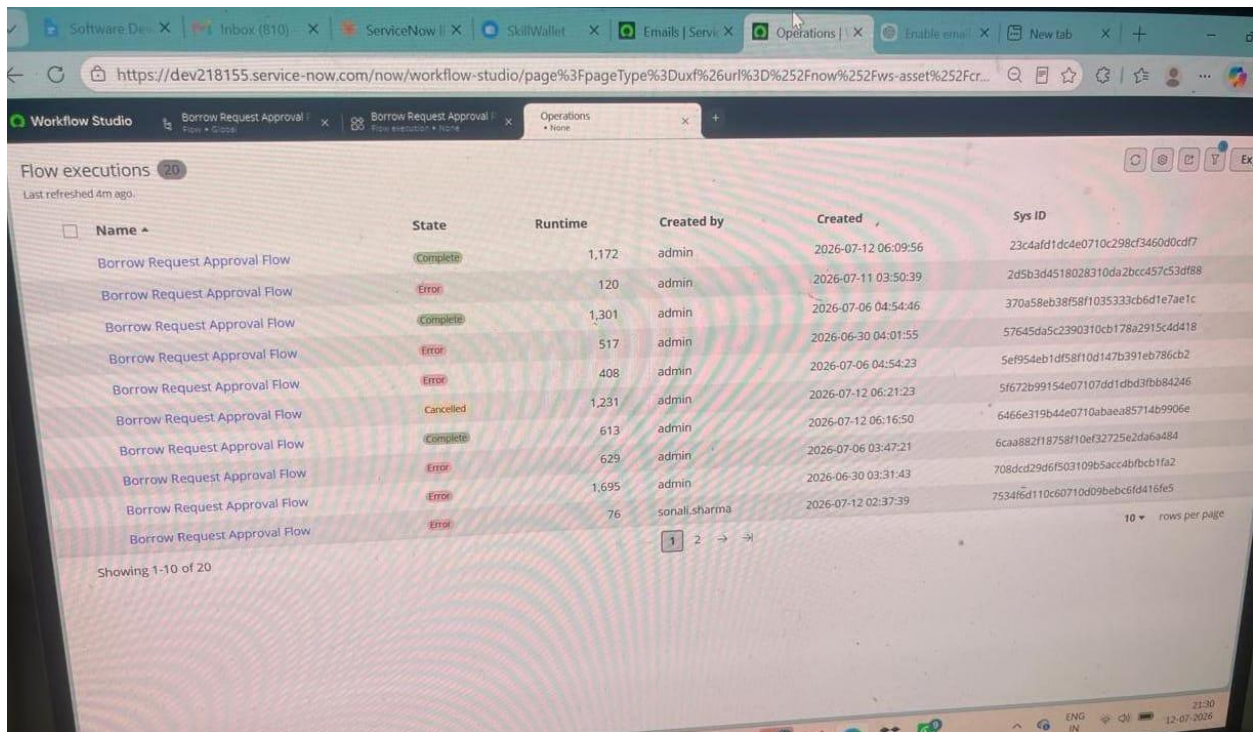
7. Results

7.1 Output Screenshots

The following visuals demonstrate the working solution (to be captured from the PDI and inserted below):

- Portal walkthrough: Home → Library Services → Submit Borrow Request.
- Live book borrow request submission by a student.
- Dynamic approval flow handled by the librarian.
- Email notification triggers for request approval and issuance.
- Request and book status tracking: Requested → Approved → Issued → Returned.
- Form UI behaviour — Return Date becomes mandatory only when a book is issued.
- Approval records visible in the Borrow Request timeline (Student request → Librarian approval → Issued).
- Flow logs showing the end-to-end execution path (Submitted → Approved → Issued → Returned).
- Most Borrowed Books and Active Book Requests reports.

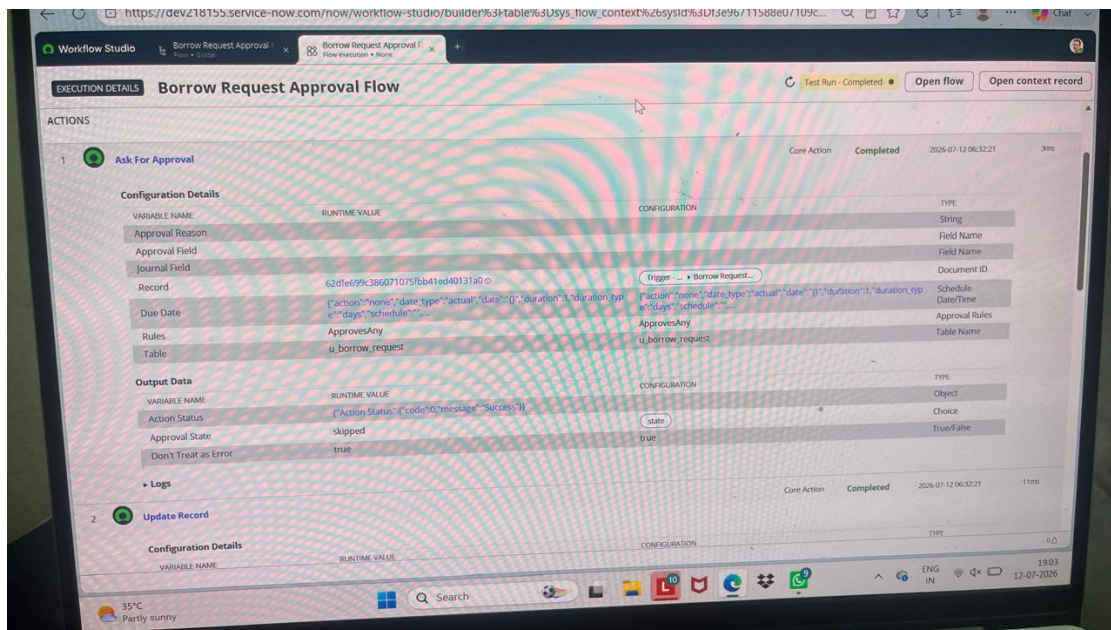
Evidence captured from PDI (screenshots below):



The screenshot displays the 'Flow executions' table in the ServiceNow Workflow Studio. The table lists 20 executions of the 'Borrow Request Approval Flow'. The columns are: Name, State, Runtime, Created by, Created, and Sys ID. The states are categorized as Complete, Error, and Cancelled. The 'Created by' column shows 'admin' for most entries and 'sonali.sharma' for one. The 'Created' column shows various timestamps. The 'Sys ID' column contains unique identifiers for each execution. The table is paginated, showing 1-10 of 20 rows.

Name	State	Runtime	Created by	Created	Sys ID
Borrow Request Approval Flow	Complete	1,172	admin	2026-07-12 06:09:56	23c4af1dc4e0710c298cf3460d0cdf7
Borrow Request Approval Flow	Error	120	admin	2026-07-11 03:50:39	2d5b3d4518028310da2bcc457c53df88
Borrow Request Approval Flow	Complete	1,301	admin	2026-07-06 04:54:46	370a58eb38f58f1035333cb6d1e7ae1c
Borrow Request Approval Flow	Error	517	admin	2026-06-30 04:01:55	57645da5c2390310cb178a2915c4d418
Borrow Request Approval Flow	Error	408	admin	2026-07-06 04:54:23	5ef954eb1df58f10d147b391eb786cb2
Borrow Request Approval Flow	Cancelled	1,231	admin	2026-07-12 06:21:23	5f672b99154e07107dd1dbd3fb84246
Borrow Request Approval Flow	Complete	613	admin	2026-07-12 06:16:50	6466e319b44e0710abae85714b9906e
Borrow Request Approval Flow	Error	629	admin	2026-07-06 03:47:21	6caa882f18758f10ef32725e2da5a484
Borrow Request Approval Flow	Error	1,695	admin	2026-06-30 03:31:43	708dcd29d6f503109b5acc4bfbcb1fa2
Borrow Request Approval Flow	Error	76	sonali.sharma	2026-07-12 02:37:39	7534f6d110c60710d09bcb6f6d415e5e

Fig 7.1 — Flow Designer execution history for "Borrow Request Approval Flow", showing multiple runs with Complete, Error, and Cancelled states captured during iterative testing and debugging.



The screenshot displays the 'EXECUTION DETAILS' for the 'Borrow Request Approval Flow'. It shows the 'Ask For Approval' action completed successfully. The 'Configuration Details' section lists variables and their runtime values. The 'Output Data' section shows the action status as 'Success'. The 'Logs' section shows the action completed at 2026-07-12 06:32:21.

VARIABLE NAME	RUNTIME VALUE	CONFIGURATION	TYPE
Approval Reason			String
Approval Field			Field Name
Journal Field			Field Name
Record	62dfe09c386071075fbb41ed40131a0	Trigger -> Borrow Request	Document ID
Due Date	{ "action": "none", "date_type": "actual", "date": "07/12/2026", "duration_type": "days", "schedule": "" }	Schedule	Date/Time
Rules	ApprovesAny	Approval Rules	Approval Rules
Table	u_borrow_request	Table Name	Table Name
Action Status	{ "Action Status": "Success" }	State	Choice
Approval State	skipped		True/False
Don't Treat as Error	true		

Fig 7.2 — Flow execution log detail: "Ask For Approval" action completed successfully (Action Status: Success), confirming the approval step of the flow executes correctly.

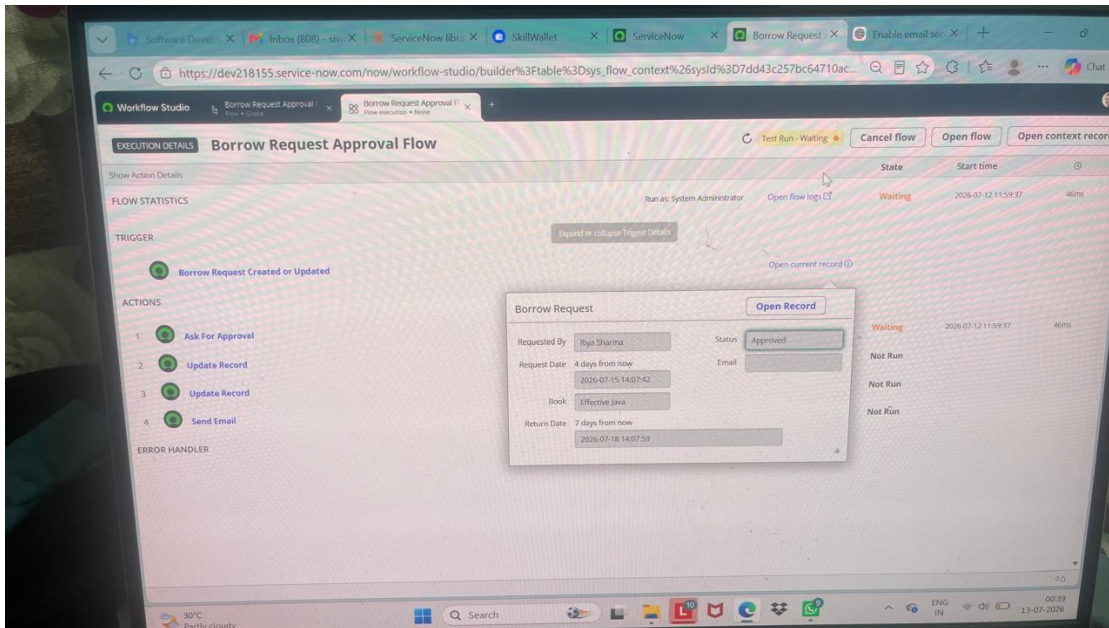


Fig 7.3 — Flow execution in "Waiting" state, correctly paused at the Ask For Approval step for a live Borrow Request (Book: Effective Java, Status: Approved), demonstrating the flow's approval hand-off to the librarian.

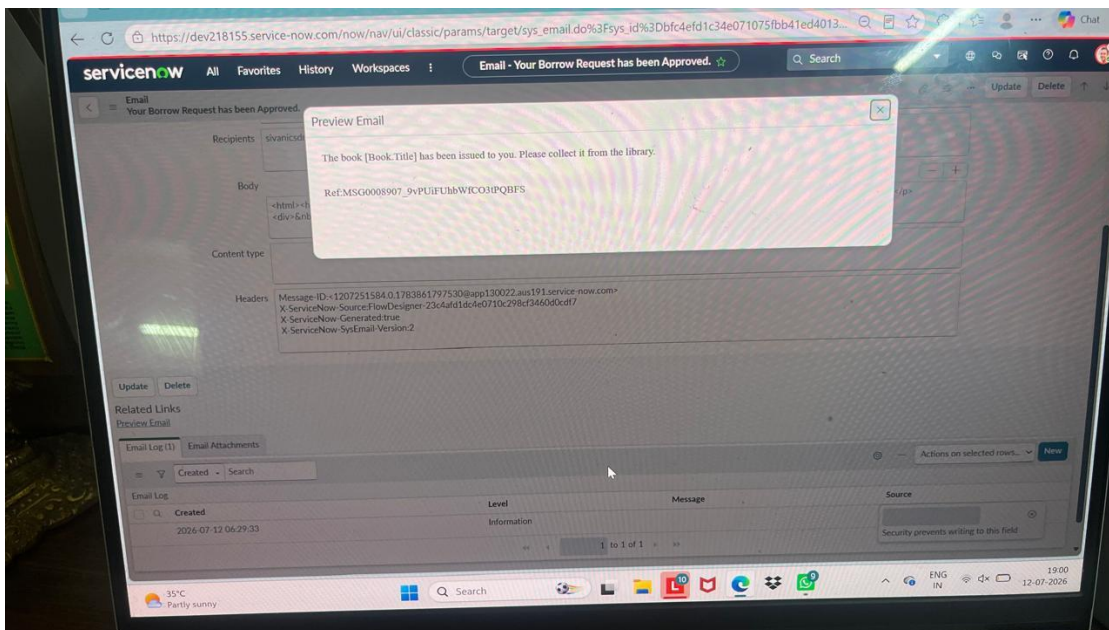


Fig 7.4 — Email notification record: "Your Borrow Request has been Approved" sent via the Flow Designer Send Email action, confirming automated email triggers for request approval/issuance.

Book	Email	Request Date	Requested By	Status
Effective Java	sivanicdmvgr00@gmail.com	2026-07-12 03:50:25	(empty)	Requested
The Alchemist	(empty)	2026-07-06 04:07:33	(empty)	Requested
Effective Java	(empty)	2026-07-15 14:07:42	Riya Sharma	Approved
First love	sivanicdmvgr00@gmail.com	2026-06-30 03:40:45	System Administrator	Requested
Effective Java	sivanicdmvgr00@gmail.com	2026-07-12 09:43:08	Riya Sharma	Approved

Fig 7.5 — Borrow Request list showing status tracking in action, including a record with Status = Approved, reflecting the Requested → Approved → Issued → Returned lifecycle.

Book - Effective Java

Title: Effective Java, Author: Joshua Bloch, ISBN: 9780134685991, Status: Issued

Email	Request Date	Requested By	Status
sivanicdmvgr00@gmail.com	2026-07-12 03:50:25	Riya Sharma	Requested
(empty)	2026-07-15 14:07:42	Riya Sharma	Approved
sivanicdmvgr00@gmail.com	2026-07-07 09:43:08	Riya Sharma	Issued

Fig 7.6 — Library Book record ("Effective Java") with Status updated to Issued, and its related Borrow Requests showing Requested, Approved, and Issued entries, confirming automatic availability updates tied to request status.

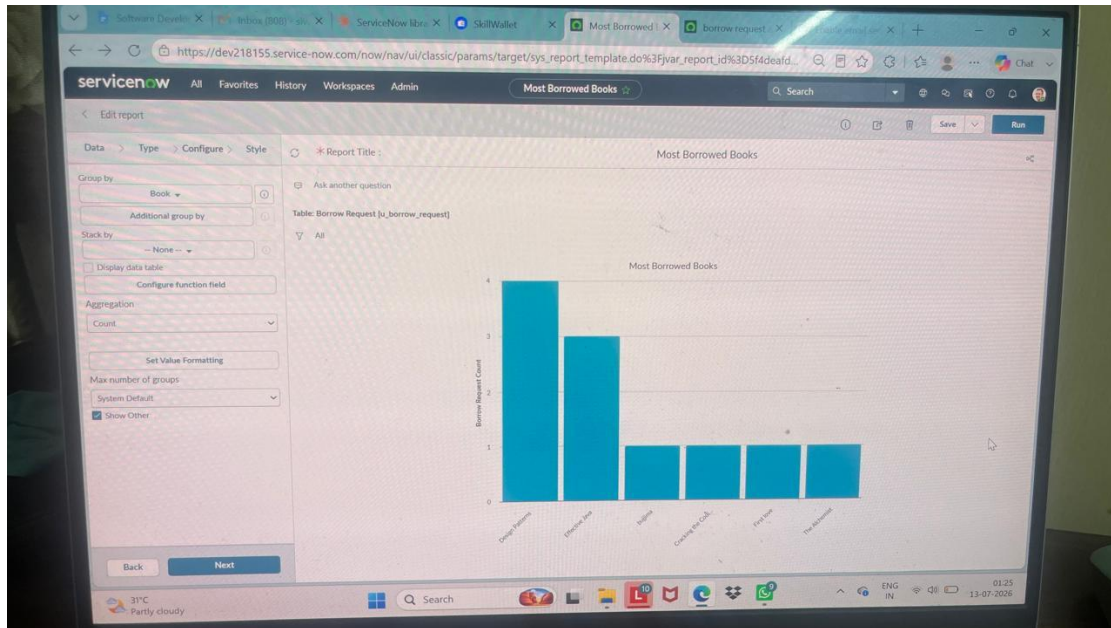


Fig 7.7 — "Most Borrowed Books" report (vertical bar chart, grouped by Book), showing Design Patterns and Effective Java as the most-requested titles — fulfilling the library analysis reporting requirement.

Book	Email	Request Date	Requested By	Status
First Love	shivanshmvg00@gmail.com	2026-06-30 02:40:45	System Administrator	Requested
Effective Java	shivanshmvg00@gmail.com	2026-07-07 09:43:08	Riya Sharma	Requested
Effective Java	shivanshmvg00@gmail.com	2026-07-12 03:50:25	Riya Sharma	Approved
Effective Java	(empty)	2026-07-15 14:07:42	Riya Sharma	Requested
Design Patterns	(empty)	2026-07-12 02:34:39	Riya Sharma	Approved
Design Patterns	shivanshmvg00@gmail.com	(empty)	Riya Sharma	Requested
Design Patterns	(empty)	2026-07-12 02:36:22	Sonal Sharma	Requested
Design Patterns	(empty)	2026-07-12 02:18:08	Riya Sharma	Approved
Cracking the Coding Interview	(empty)	2026-07-13 12:35:04	Riya Sharma	Approved
Cracking the Coding Interview	(empty)	2026-07-14 03:03:49	Riya Sharma	Rejected

Fig 7.8 — "Active Book Requests" view: Borrow Request list filtered by Requested By Active = true, showing all 10 in-scope requests across Requested, Approved, and Rejected statuses — fulfilling the Active Book Requests reporting requirement.

8. Advantages & Disadvantages

Advantages:

- Removes manual register-based tracking; all data is centralized and auditable.
- Automated approval and status updates reduce turnaround time for students.

- Role-based ACLs ensure data integrity and appropriate access boundaries.
- Built-in reporting gives librarians visibility into borrowing trends without extra tooling.

Disadvantages:

- Dependent on ServiceNow platform availability and licensing for institution-wide rollout.
- Current scope does not yet include automated overdue reminders (planned as future enhancement).
- No integration yet with external library catalogs or digital resource systems.

9. Conclusion

The Library Management Project enhances book borrowing and administration in ServiceNow through structured roles, automated approval flows, and workflow-driven processing. Students are able to request books seamlessly, while librarians manage approvals and issuance with built-in validations. Clear status updates and reporting improve visibility and control, and the automated design ensures efficiency, reliability, and scalability for future academic service extensions.

10. Future Scope

- Auto-notifications for overdue books and reminders via email/SMS.
- Integration with external library systems or digital catalog APIs.
- Enhanced reporting dashboards covering borrow frequency, student activity, and overdue analysis.
- Expansion to other academic services such as lab equipment requests, study room booking, and digital resource access.

11. Appendix

GitHub & Project Demo Link:

Links would be provided in skill wallet.